

Virtual forests for decision support and stakeholder communication

Stefan Holm^{*}, Janine Schweier

Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Zürcherstrasse 111, CH-8903, Birmensdorf, Switzerland

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ABSTRACT

Challenges in forest management are increasing due to climate change and its associated risks. Considering the needs and demands of various stakeholders leads to more complex decision-making. The increasing amount and quality of available geographic, forest and individual tree data, the combination of this data, and the use of forest growth simulators make it possible to support forest managers in this decision-making process. Our aim was to develop a strong visualization instrument that can be used in both forest planning and stakeholder communication. We present a solution based on a game engine, where data from multiple sources (terrain data, satellite imagery, tree data) is combined into a virtual environment. The user can move freely inside this virtual forest, look at the forest from arbitrary perspectives, and observe its development over the years under different management scenarios. We demonstrate the usefulness of this approach with a study region in Switzerland.

1. Introduction

1.1. Increasing challenges in forestry

Climate change is currently strongly affecting the environment worldwide (FAO, 2013), including forests, which cover approximately one-third of the global land area (Forest Europe 2019). Forests are particularly vulnerable, due to expected increases in droughts and in the intensity and frequency of climate-change-induced disturbances (Forzieri et al., 2021; Anderegg et al., 2022). The resulting risks and uncertainties are – for various reasons – a large challenge for forest managers on strategic, operational, environmental and financial levels (Sanginés de Cárcer et al., 2021).

To tackle climate change and its consequences, many driving factors for resilient forest management have been identified (Aggestam et al., 2020). These include diverse management approaches currently applied in Central Europe, such as integrated forest management, continuous cover forestry, close- and closer-to-nature management, and climate-smart forestry (Seidl and Lexer, 2013; Brang et al., 2014; Jandl et al., 2018; Krumm et al., 2020; Mason et al., 2022). The aim of these approaches is to increase the resilience of future forests and to reduce and/or avoid ecosystem damage and the social and economic effects caused by climate change. Resilient and diverse managed forests are considered to be able to maintain and enhance the provision of biodiversity and multiple forest ecosystem services, including provisioning (e.g. timber, protection), regulating (e.g. carbon storage), supporting (e.

g. nutrient cycles), and cultural (e.g. recreation) services.

As a consequence, forest structures are being developed towards more mixed forests (e.g. increasing shares of mixed forests), silvicultural strategies are being adapted (e.g. continuous cover forestry, as acknowledged in the European Union Forest Strategy, EC, 2021), and sustainable forest operations are being applied (e.g. use of cable yarders for extraction in flat terrain to protect soils (Schweier et al., 2022)). However, even though the adaption towards more resilient forests has started, there are shortcomings (Brang et al., 2014) due to lack of experience and the rapid speed at which climate change is happening.

Thus, we argue that new tools need to be developed to better support forest managers. In view of the complexity of the subject, the multi-objective forest management aims that are pursued, and the limited experience in adaptive forest management combined with a growing share of uncertainty, the increasing complexity in decision-making for forest managers must tackle needs to be addressed. This becomes even more evident when considering the time horizons forest managers have to deal with: the average rotation period of a forest is, depending on the tree species and site conditions, much longer than a human generation.

While current research is much focused on the development of new silvicultural management approaches and the inclusion of risks and uncertainties, there are numerous recent innovations in digital transformation at the same time. However, to our best knowledge, there is no IT-application yet that combines both domains. A powerful instrument visualizing potential impacts of different management options as realistic as possible is missing, but would be of high value for forest

^{*} Corresponding author.

E-mail addresses: stefan.holm@wsl.ch (S. Holm), janine.schweier@wsl.ch (J. Schweier).

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managers, either to support their planning or their communication with stakeholders.

1.2. Use of data in forest management

Forest growth simulators, such as *ForClim* (Bugmann, 1996), *Waldplaner* (Hansen and Nagel, 2014) and *SwissStandSim* (Zell et al., 2020), are valuable instruments to estimate the future development of forests depending on the applied management strategy. To make use of them is of high importance when considering the above-mentioned challenges. Further, simulation outputs can be connected with indicator value functions to estimate the future provision of biodiversity and ecosystem services, as done, for example, in Blattert et al. (2020) and Thrippleton et al. (2021). To enhance and facilitate this important step of forest planning, it would be of high value to visualize those simulation outputs. Some growth simulators provide simple perspective stand views and crown charts (e.g. SILVA, Pretzsch et al., 2002). However, those illustrations of trees are based on simplified and rudimentary geometric forms like cylinders and cones. Some simulators offer more features, e.g. the Swedish Heureka contains a suite of applications such as StandWise - a stand-level management simulator including visualization in 2D and 3D (Wikström et al., 2011). However, tree forms are still relatively simple and the visualizations do not contain ground vegetation or other visual representations of the surrounding landscape. To conclude, to our best knowledge, modern and immersive 3D visualizations making use of today's state of the art in computer graphics are not yet existing.

Due to recent technological progress in hardware and software, a large amount of data is available and often free of charge. Most countries conduct national forest inventories (e.g. Gschwantner et al., 2022) on a regular basis. This has become possible through the increasing use of remote sensing methods (Fassnacht et al., 2023). Moreover, many forest management units collect their own high-resolution data. Technological developments facilitate the processing of large amounts of data (Bartodziej, 2017), such as Light Detection and Ranging scans (LiDAR), which are collected with aerial laser scanners (ALS) and/or ground-based LiDAR, such as terrestrial or mobile laser scanners (TLS, MLS; Murtiyoso et al., 2023; Zürcher et al., 2023). An example is the "3D Forest" tool (<https://www.3dforest.eu/>) for point cloud data processing from forest environment acquired by terrestrial laser scanner or research activities such as the COST action 3DForEcoTech (<https://3dforecotech.eu>).

Having access to such spatially explicit information on the single tree level is not yet given in most cases but would significantly enhance the quality of data used by forest planners. The most common variables are probably tree species and diameter at breast height (DBH), but there are more variables on which forest practitioners base their management, such as social structure, tree structure, crown ratio, crown volume, merchantable volume, existence of nest holes, stem defects, bends and more (Griess and Schweier, 2023). Moreover, many other types of environmental data are available; in the case of Switzerland, this includes digital height models (terrain and surface), orthoimages, and road network data, to name just a few relevant examples. None of the existing forest management tools integrates this open data.

1.3. Forest visualization as a potential solution

It is therefore almost imperative to make use of spatially explicit data to better support forest managers in the complex process of decision making. However, the handling of such a large amount of data is a challenge. To solve this problem, data could be connected in a virtual environment, similar to how it is done in other disciplines (e.g. landscape planning; Orland et al., 2001). This would make it possible to recognize, in only a short time, many aspects that are relevant to decision-making. Moreover, modern immersive visualization technologies, such as augmented reality (AR), virtual reality (VR), and mixed reality (MR) (Murtiyoso et al., 2023) easily allow the user to change

perspectives, e.g. to zoom into or out of a forest stand, and thereby assess aspects that are important on the tree, stand or landscape level.

Besides addressing the complexity involved in decision-making, such visualizations would be of particular interest in supporting stakeholder dialogs. The public is taking an increasing interest in forests. It has been proven that forests contribute to human well-being (Bratman et al., 2012; Forest Europe, 2019; Marta-Pedroso et al., 2014), and people are interested in what is happening in the forests. However, the public usually does not have a professional forestry background, which is leading to increasingly frequent conflicts (Nousiainen and Mola-Yudego, 2022). In Niemelä et al. (2005), increasing recreational needs, an increased importance of the environmental movement, and an intensification of forestry operations are mentioned as the main drivers of conflicts. The last of these factors was also analyzed by Dög (2013), who conducted a comprehensive survey (N = 491) of forest visitors and found a generally negative attitude towards logging machines in the forest among almost half of the forest visitors surveyed. Hellström (2001) states that conflict management focusing on improving the relations among the conflict actors leads to milder conflicts. It can be concluded that it is worth investing efforts in relation management involving all forest stakeholders.

In this study, we therefore aimed to develop a strong visualization instrument that can be used in both forest planning and stakeholder communication. This virtual forest approach is new, and no such application exists yet.

1.4. Virtual forests: history, definition and applications

Computer-based visualizations of forests have a long history in forestry, often with the stated goal to demonstrate consequences of different forest management strategies (Murtiyoso et al., 2023). The first attempts were reported at the end the 1980s, when individual forest images were manipulated "to speculate about the visible consequences of different forest management practices" (Orland, 1988). In the 1990s, there was a shift from manipulated photographs to computer-generated three-dimensional (3D) images (Bergen et al., 1998; Buckley et al., 1998). In the first decade of the 21st century, there was another shift from single images to animations, yet they were still restricted to pre-defined camera angles (Dunbar et al., 2004; Wang et al., 2006). Forest visualizations subsequently became faster and thereby made it possible to freely move around inside the visualized forests (Pretzsch et al., 2008; Cournède et al., 2009), with some systems using multiple screens to extend the field of view (Stock and Bishop, 2006; Boukherroub et al., 2018; Fujisaki et al., 2008; Fabrika et al., 2018; Fabrika, 2021). In the last years, head-mounted displays have been used to enhance the immersiveness (Astner, 2018; Holopainen et al., 2020; Botev and Viegas, 2020; Chandler et al., 2022; Huang et al., 2021).

Murtiyoso et al. (2023) define virtual forest as "3D representations of forests that can be used for a variety of applications, such as research, training, gaming, and forest planning". This definition is used here as well. They describe three areas of application for virtual forests in detail: virtual forests as a tool for education and training, as a tool for forest management, and as a tool for communication among stakeholders.

Some studies focus on the impact of virtual environments in supporting relaxation and meditative processes in humans, such as Hejtmánek et al. (2022) and Reese et al. (2022). Others concentrate on the advantages of 360° virtual forest tours in academic forestry education (Foehrder et al., 2021). Virtual or augmented environments are also used to better understand human-nature relationships and behavior patterns (Nitoslawski et al., 2021). What is missing is an application which supports forest managers in planning and communication activities by visualizing the potential consequences of alternative silvicultural management strategies.

2. Materials and methods

2.1. Game engine

The core of our virtual forest application is a game engine. A game engine is a software platform that facilitates the development of a virtual world with which a user can interact. Such virtual worlds are often adventure games, first person shooters, or car racing games. In our case, this virtual world is a digital twin of an existing forest. The use of a game engine makes it possible to focus the development effort on the visual fidelity and the user interaction, while the game engine handles a broad range of functionality autonomously in the background. The typical functionality of a game engine includes the handling of light (e.g. directional light, ambient light), physics (e.g. collisions of objects), sound, network, level of detail, and user input on various devices (e.g. desktop computer or game console).

In a first prototype, we used JMonkeyEngine (<https://jmonkeyengine.org/>) as the game engine. We later switched to Unity (<https://unity.com/>) due to its comprehensive range of ready-to-use extensions (“assets”) that simplify the development process. However, the use of proprietary assets forbids making the whole application open-source.

With some additional assets from the Unity asset store, we configured the game engine to simulate different weather conditions and realistic positions of the sun and the moon throughout the year and throughout the day for the location of the visualized forest.

2.2. Input data

To create a virtual world with a game engine, it needs to be defined how this virtual world should look like. In the given use case, this means that the surface of the world (the terrain) and the objects which are placed on this surface (trees, shrubs) need to be defined. The data used for this purpose is presented in detail in the following sections.

2.2.1. Terrain data

The terrain mesh forms the ground of the virtual world; it is the 3D structure on which the player can walk. The terrain mesh is a combination of the digital elevation model (DEM) swissALTI3D (Swisstopo, 2020) and the surface model swissSURFACE3D (Swisstopo, 2018), both of which have a resolution of 0.5 m (Fig. 1, top images). The core study region, i.e. the forest area where the 3D models of the trees are placed, uses the DEM. The rest of the 1×1 km area uses the surface model, so that forests outside of the core study region, and also bridges and buildings, are displayed with their real-world height. The height data is stored in a grayscale image with a resolution of 16 bits per pixel, allowing 65,536 grades of height (instead of just 256 grades with 8 bits per pixel, which would have led to a terrace effect).

2.2.2. Satellite imagery and ground textures

The texture placed on top of the terrain mesh is a combination of different images. The base texture is an orthoimage of aerial photographs with a ground resolution of 0.1 m (Swisstopo, 2022) and is used outside the core study region. The goal of this texture is to visually embed the core study region within its real-world surroundings. The

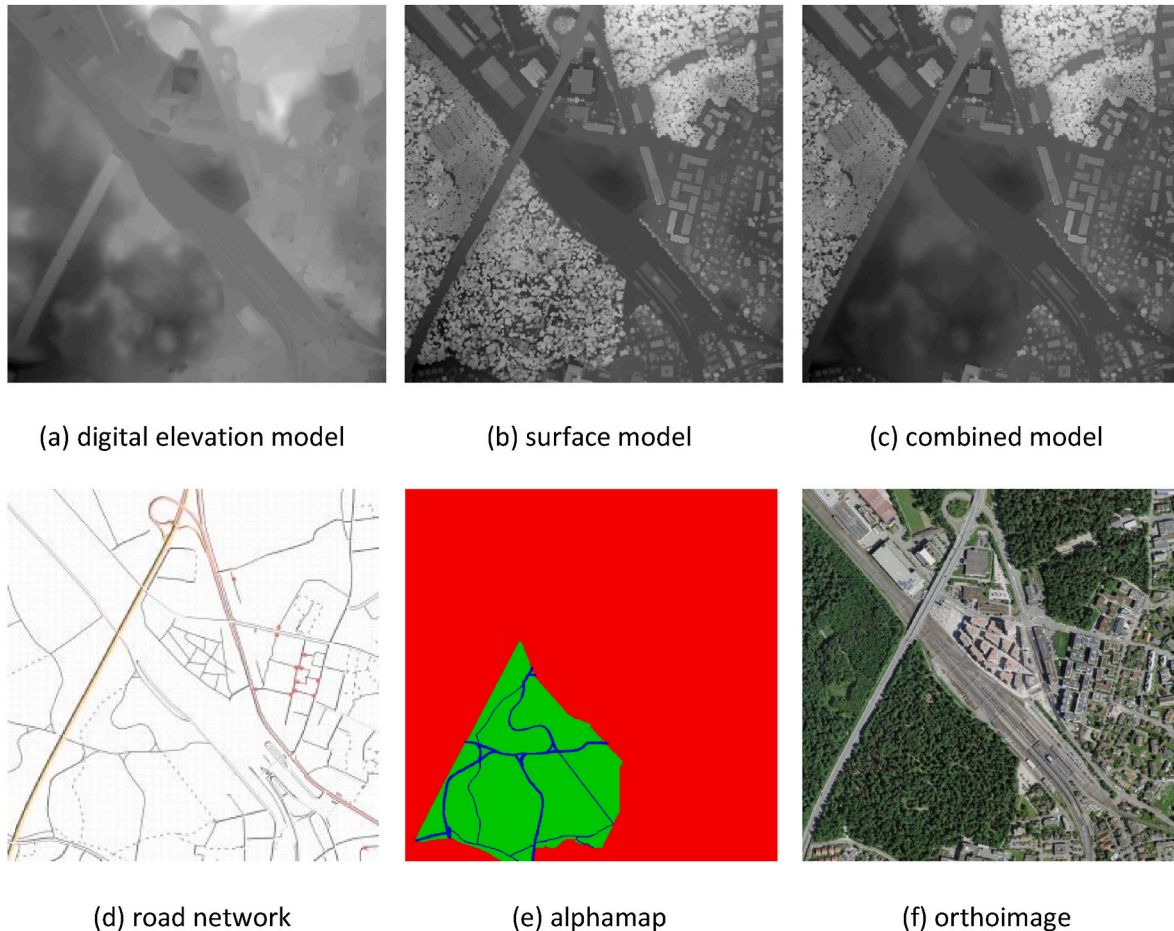


Fig. 1. (a) Digital elevation model swissALTI3D, (b) surface model swissSURFACE3D, (c) combined elevation/surface model, (d) road network, (e) alphamap, and (f) orthoimage.

core study region uses patches of two additional textures, a generic road texture, and a generic forest ground texture. The regions where each of these textures is applied to are defined in an alphamap, where blue stands for roads (based on the road network data set “swissTLM3D road and tracks”; *Swisstopo*, 2023), green stands for forest, and red stands for orthoimage texture (Fig. 1, bottom images).

2.2.3. Tree data

In addition to the data used for the terrain geometry and appearance, data about the location and the properties of the trees in the core study region is needed. This data can be either static, e.g. data from a marteloscope, or dynamic, i.e. data from a forest growth simulator. In the present study, we used data generated by different forest growth simulators.

2.2.4. Integration of the different data sources

When integrating the data presented in the previous sections, important points that need to be considered are the coordinate system and, related to that, the geographical boundaries of the input files. Concerning the latter, it is important that the images containing the elevation data and the satellite image have the same geographical boundaries, to ensure that every single point on the resulting terrain reflects the correct elevation and color of its real-world counterpart and there is no horizontal displacement between elevation and color when they are overlaid to form the 3D-landscape.

Particularly the coordinate system is important when tree position data are combined with the terrain. The tree positions specified in a marteloscope or a forest growth simulator might be present in a local coordinate system, however, the coordinates of such a local coordinate system need to be converted to the coordinate system of the terrain or vice versa. Finally, these coordinates have to be converted to the coordinate system of the game engine.

2.3. Forest growth simulators

To make this application useable for various cases, we implemented an importer tool that allows importing tree data from (currently) three forest growth simulators that are used in Central Europe, namely *SwissStandSim* (Zell, 2016; Zell et al., 2020), *Waldplaner* (Hansen, 2012), and *ForClim* (Bugmann, 1996; Rasche et al., 2012). We have chosen those simulators because they were developed for Central European conditions, are commonly used and we are familiar with them. The idea, however, is to allow importing tree data from additional simulators in future. The importer tool takes the output files generated by the forest growth simulators (or in the case of *Waldplaner*, connects to its internal database) and converts this data into a common format that can later be processed by the virtual forest application. For each scenario simulated by one of the forest growth simulators, two csv files are generated. The first file includes all static data about the forest, i.e. the position of the trees, their species, the year of their appearance, and the year of their disappearance. The second file includes all dynamic data, such as tree height, DBH, volume, and crown base and width for each year during which the tree is standing and for which the forest growth simulator calculated values.

The three forest growth simulators have some similarities and differences. Common to all is that they differ between different tree species and calculate height, DBH and volume for the trees in a given forest patch for a sequence of years under different scenarios. However, only *Waldplaner* uses explicit coordinates for each tree inside the patch. For the two other simulators, the exact coordinates within the simulated patch are not known, only the coordinates of the simulated patch itself. And even from the data of *Waldplaner*, the explicit coordinates are only known for the trees within the simulated patches, not for the trees between the patches. This creates a problem common to all three forest growth simulators: how can we determine the location and distribution of trees between the simulated patches, and how can we determine the

location of the trees inside the patches, when this information is not already known?

The algorithms the three forest growth simulators use to solve this problem are all based on the same idea. For a piece of forest where the exact coordinates of the trees are not known, the trees are placed randomly, with the tree density and tree distribution taken from the closest patch. The tree density can be calculated for all three forest growth simulators by dividing the area of the simulated patch by the number of trees inside the simulated patch. Consequently, for *Waldplaner* this algorithm is used only for the forest area between the simulated patches, while for *SwissStandSim* and *ForClim* it is used for the forest areas both inside the simulated patches and between the simulated patches. This algorithm makes it possible to populate the complete forest area with trees.

2.4. 3D models of trees

With the data about the location, species and dimensions of the trees standing in the forest at hand, as the next step we needed 3D models of these trees that can be placed inside our virtual world. A tree model consists of its 3D structure and one or more textures.

To generate the 3D structure of the tree models, we used the software *Speedtree* (<https://store.speedtree.com/>). This software enables the procedural generation of arbitrary-looking trees by defining parameters for the appearance of the trunk, branches and leaves. Parameters such as tree height, DBH, stem form, height of the crown base, angles between branches, and density of branches and leaves are used to produce realistic models of existing tree species. Procedural generation also means that the generated trees could easily be randomized so that all generated trees look slightly different. In our case, for each species present in the forest of our study region, we modeled three trees of different sizes. With a random rotation around the vertical axis of the tree when the tree was placed in the virtual forest, this gave enough visual heterogeneity between the trees so that they would not be perceived as equal tree models. The sizes were chosen such that the resulting trees could be scaled with a factor that was not too large to distort the tree in a visually disturbing way. As a rule of thumb, we chose the heights of the three tree models to be around 7 m, 14 m, and 28 m. We diverged from this rule to account for the actual distribution of tree dimensions of a given species in our study region. This led to the final selection of trees modeled for this project (Table 1). In the current version, the 10 most important tree species of Switzerland and its neighboring countries are implemented.

To make the tree models look realistic, they not only need to have a 3D structure that reflects the structure of a real tree, but also need to have realistic-looking textures. Textures are image files that are placed on top of the 3D structure of the tree model. They represent the appearance of the bark and the leaves or needles. Different texture types serve different purposes: the basic texture is the albedo texture, which contains the color information for the given surface. Additional textures used in this project were normal textures, which contain information about the surface structure; opacity textures, which define transparent regions, e.g. around or within leaves; and gloss textures, which define specular light reflection properties of the surface. This concept of rendering objects similarly to how light and surfaces interact in reality is called physically based rendering (Pharr et al., 2016). The textures were bought on respective online stores, many of them from Xfrog (<https://www.xfrog.net/>), who also offer very detailed 3D models of trees. However, their tree models consist of 50,000 to 500,000 polygons per tree, because they usually model each leaf with an individual polygon. This leads to high visual quality, but also to high performance costs, so that for the given use case these models cannot be rendered fast enough in real-time in sufficient quantity on average graphics cards.

The tree models we created usually consisted of around 800 to 1000 polygons per tree. To achieve such low polygon counts, we also used *Speedtree* to create 3D models of twigs that contain multiple leaves, around 100 to 300 per twig. This twig model was then exported as a

Table 1

Overview of the modeled trees currently available in the virtual forest application. Tree species reflect the most important species in Switzerland and its neighboring countries. DBH = diameter at breast height.

Species	height [m]/DBH [cm]/crown base [% of height]/crown width [m]											
	Tree 1 (small)				Tree 2 (medium)				Tree 3 (large)			
<i>Abies alba</i>	6	7	60	2.8	15	14	60	3.4	31	35	60	5.8
<i>Acer pseudoplatanus</i>	4	4	50	3.2	18	12	60	4.1	28	34	60	7.0
<i>Alnus incana</i>	6	6	50	1.5	15	11	50	2.4	25	28	50	5.2
<i>Betula pendula</i>	7	7	50	2.1	17	11	50	2.5	26	29	50	5.2
<i>Fagus sylvatica</i>	4	4	50	2.5	14	12	50	3.5	30	35	50	7.6
<i>Fraxinus excelsior</i>	5	3	50	1.7	18	10	70	3.2	29	36	60	7.6
<i>Larix decidua</i>	5	7	50	0.8	15	12	70	1.9	30	37	70	5.9
<i>Picea abies</i>	5	7	50	2.2	15	13	65	2.8	30	35	65	5.2
<i>Pinus sylvestris</i>	5	6	50	1.2	13	13	70	2.2	31	40	75	6.2
<i>Quercus petraea</i>	5	4	50	2.0	15	10	70	3.0	30	37	70	7.2

texture image and applied to the 3D model of the whole tree. Using this approach, we did not need a single polygon per leaf, but only two to four polygons per twig, which reduced the polygon count of the tree model significantly.

Finally, we considered different wind settings to visualize the forests as naturally as possible. Speedtree simplifies the definition of wind settings and the different levels of detail (LOD) for the trees. In the wind settings, it can be defined how strongly different parts of the tree react to different strengths of wind. Thus, the user does not see a static image on the screen, but one with trees slightly blowing in the wind. LODs are a means to reduce the polygon count of the rendered scene. Trees close to the player are rendered in full detail, while trees far away from the player are rendered with fewer polygons. We defined three different LODs for each tree, which seemed a good balance between the possible number of polygons that can be saved and the additional number of 3D models that have to be kept in the memory. A comparison of Speedtree and other 3D modeling software for trees and plants can be found in Romeijn et al. (2012).

Fig. 2 shows an example of a tree model we created and how the adjustment of parameters, such as the position of the crown base or branch density, influences the visual appearance of the tree.

2.5. Study region

In principle, the application could be applied to visualize the future development of any simulated forest. To demonstrate this, we applied the virtual forest application in a study region located in the city of Bülach in northern Switzerland. This study region was selected for two reasons. First, it was the study region in two previous research projects, and therefore we had existing output data from the simulations of the forest development being conducted with two forest growth simulators (*SwissStandSim* and *Waldplaner*). Second, it is an interesting case for stakeholder communication. The depicted forest is enclosed by railways, a highway, and the city (Fig. 3). Many people seek recreation in the area, and there is a fitness parkour route, a bike trial, and an educational trail concerning middle spotted woodpeckers (*Dendrocoptes medius*). The area

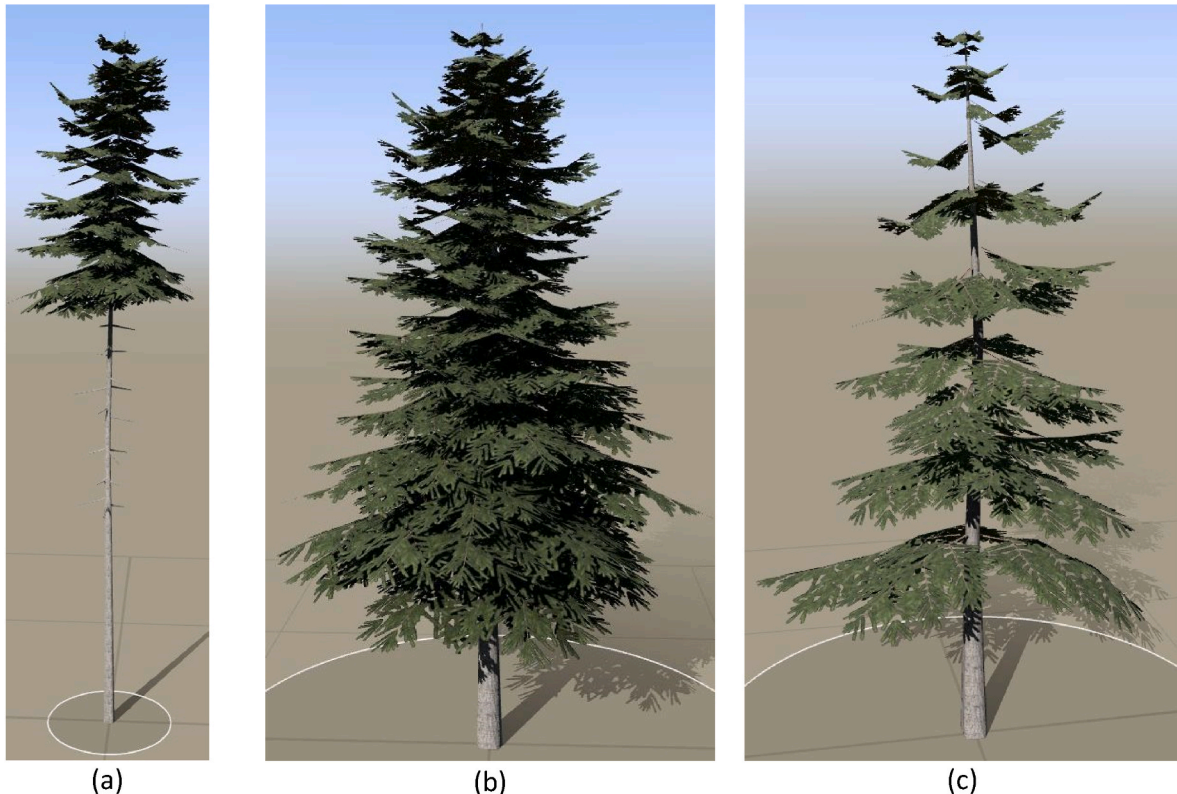


Fig. 2. A single model of a Norway spruce (*Picea abies*) in three variations: (a) high crown base, (b) low crown base, and (c) sparse crown.



Fig. 3. The study region: (a) location of Switzerland in Europe (image from openstreetmap), (b) location of study region in Switzerland (image from geo.admin.ch), (c) orthoimage of the study region in Bülach (image from geo.admin.ch), and (d) aerial image (own image).

is important from the point of view of nature conservation. It consists of oak-rich forests, and around one-fifth of the forest is oak nursery land. Apart from the oak nursery areas, the forest is managed according to the principle of continuous cover forestry. In the study area, there are various tree species, with sessile oak (*Quercus petraea*) and European beech (*Fagus sylvatica*) as the main species (Fig. 4).

The area visualized in the application is 1×1 km (including the surroundings of the forest), while the visualized forest itself is approximately 19 ha.

3. Results and discussion

The virtual forest application makes it possible to look at the represented forest from any perspective. This is implemented for two modes of movement: a walking mode (Fig. 5) and a flying mode (Fig. 6). The movement is controlled by the standard WASD keys used in many games, i.e., W for forward movement, A for left movement, S for backward movement, D for right movement. The mouse is used to control the viewing direction. Some additional keyboard keys can be used to change the time of the day, the current year, the weather conditions, or to switch between walking and flying mode.

3.1. Visual perception and accuracy

The basic prerequisite for fluent real-time movement in the virtual forest is that the application is fast enough to render continuously and, at any time, with a sufficient number of frames (i.e. individual images) per second (FPS). Which exact number of FPS is considered sufficient is controversial (Brand, 2001; Hagström, 2015; Wilcox et al., 2015; Paz-hoochi and Kingstone, 2021) and also depends on the type of visualized content and its interactivity, i.e. whether it is a movie or a game, and in the latter case, what kind of game. Read and Meyer, 2000 state that the human eye can perceive 10–12 images per second as individual images, while more than 15 images in a second will “create the sensation of visual continuity” (Read and Meyer, 2000). Cinema movies traditionally have 24 FPS (Wheeler, 2012; Wilcox et al., 2015). On game consoles, 30 FPS was the standard for a long time, but today they mostly run at 60 FPS (Gapo, 2022). In interactive games, especially shooters and multi-player games, a frame rate of 60, 120 or even more FPS can be a competitive advantage for the player (Hagström, 2015; Gapo, 2022). However, in our application, where there is no interaction with other players and the only purpose of movement is to change the viewpoint on the forest, a frame rate never falling below 30 FPS is considered sufficient. In our application, the frame rate when flying over the forest is slightly lower than when walking through it, but the value always stays above 30 FPS with a standard desktop machine with a mid-range

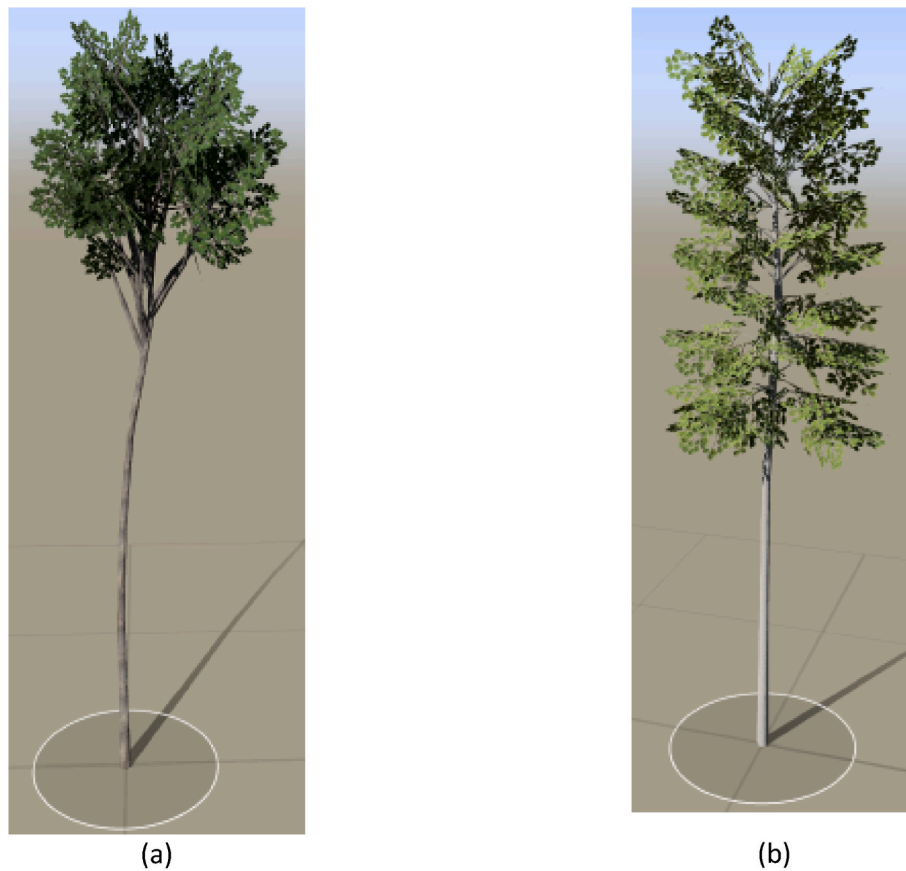
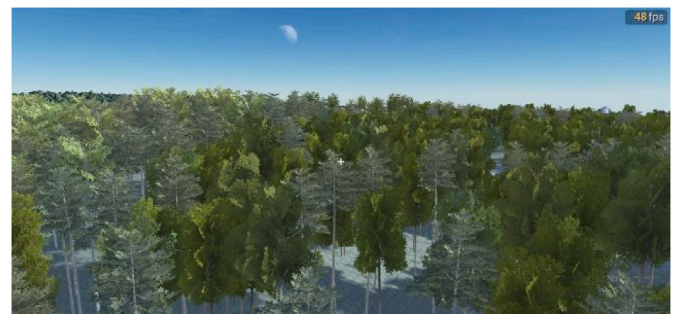


Fig. 4. The two main species in the study region: (a) sessile oak (*Quercus petraea*) and (b) European beech (*Fagus sylvatica*).

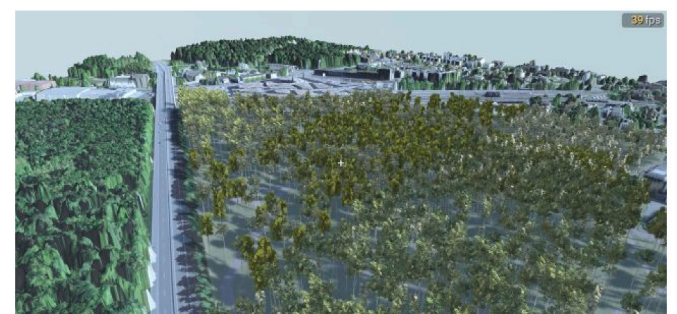


Fig. 5. Screenshot taken from the visualization simulating walking through the virtual forest.

graphics card (the application was tested on several different machines containing different graphics cards each with a price around USD 200). The frame rate is influenced by many factors. A major influence on the frame rate is the number of polygons in the visualized scene. In a virtual forest application, the number of polygons is mainly determined by the number of trees and ground vegetation plants and by the number of polygons per tree or other plant. As the number of trees and ground vegetation plants is given by the situation in the study region, the key to reducing the number of polygons in the scene is to reduce the number of polygons per tree or plant. However, the number of polygons per tree correlates with its visual accuracy, i.e. reducing the number of polygons of a tree also decreases its visual fidelity. There are some tricks to reduce the number of polygons substantially while only slightly reducing the visual quality. An important one is, as described in section 2.4, using twig textures that comprise many leaves in a single image instead of modeling every leaf individually. For this application, we tried to keep



(a)



(b)

Fig. 6. Screenshots taken from the visualization simulating flying over the virtual forest at low height (a) and elevated height (b). In (b), the study region is located on the right side of the highway, where all trees are modeled individually. On the left side of the highway, the forest is based on an orthoimage and a digital elevation model.

the models of all trees below 1500 polygons, and many of the trees are around 800–1000 polygons in the version with the highest level of detail. The second LOD usually has around half the polygon count compared with the most detailed LOD, and the third LOD around a quarter of it. Due to their physical structure, conifer trees are often easier to represent with fewer polygons than broadleaves.

The largest visual accuracy of the trees in a forest would be achievable by using point cloud data from LiDAR measurements. Two major problems come with this approach: first, the point clouds need to be converted and optimized to 3D models with a sufficiently low polygon count. The conversion of point cloud data to meshes has been discussed previously (Bonnafe et al., 2007; Wang and Chu, 2009; Laksono and Aditya, 2019). Second, this approach only improves the visual accuracy of the current situation in the forest; to visualize the future development of the forest, there is obviously no possibility to conduct LiDAR measurements and obtain the corresponding point clouds. For these reasons, we chose to stay with a few generic tree models for each species and scale them to their real-world height and DBH, which is faster but may not be as visually accurate as tree models generated from LiDAR measurements.

3.2. Visual data analysis

Forest growth simulators allow to calculate the development of the trees in a forest based on environmental or management scenarios for a given range of years. To visualize this temporal scale of the data, we implemented a time-travel function. This allows the user of the virtual forest application to travel forwards and backwards in time while staying at an arbitrary location inside or over the forest. This offers a novel method to analyze the simulated scenarios, in addition to classical statistical analysis of the generated data.

This time-travel function has great potential to support environmental and social impact assessments. In their daily work, forest managers make long-lasting decisions benefiting future generations, while at the same time meeting expectations and needs for various ecosystem services of today's generations. Many forest management approaches are currently applied to increase the resilience of future forests, however, yet there is limited experience in adaptive forest management combined with a growing share of uncertainty. It might be a very useful function to visualize the potential positive and negative effects of different management scenarios today and in the future before they are implemented to allow decision-makers informed decisions to be made.

In addition to the possibility of time-travel between different years, it is also possible to time-travel between different days throughout the year and different times of the day. As the sun position throughout the day and the year inside the application reflects the real sun position for the given region and time, the application therefore allows the user to observe the amount of light and shadowing of the forest ground at any position at any time, which is an important influencing factor for the forest regeneration (Knüsel et al., 2017; Kanjevac et al., 2021) as well as for other functions, e.g. outdoor routes or areas with cooler, shaded conditions could be identified for recreationists.

3.3. Ground vegetation

The visualization of the forest ground and its vegetation has a large influence on how realistic the visualized forest is perceived by the user of the application. However, two problems are related to this: first, there is no data about the exact species and location of the ground vegetation. Second, the rendering of a myriad of small plants has a large negative effect on the frame rate and therefore on the overall experience of the user. We therefore tried to find a balance between rendering too few ground vegetation models (and thereby losing visual fidelity) and rendering too many ground vegetation models (and thereby losing performance).

3.4. Scale

The forest represented in this project has an area of approximately 19 ha and is embedded in a visualized area of 1 km². In the literature, forest visualizations exist at different scales. Dunbar et al. (2004) differentiate between the plot level, stand level and landscape level. Song et al. (2006) distinguish between the stand scale, near-landscape scale, and regional landscape scale. Pretzsch et al. (2008) differentiate mainly between the single tree, stand and landscape scales, but also mention the enterprise, regional and country scales. In the present application, the visualization can be continuously zoomed in or out of the forest, to look at individual trees in detail or at the forest as a whole.

3.5. Use of the application

The application of such virtual forests could be of high value for forest planners by visualizing the long-term consequences of management decisions. Users can adopt different perspectives and thereby expand their horizons. Moreover, a wide range of different data can be integrated, leading to more informed decisions. The application seems to also be well suited for education and training purposes, as described in Murtiyoso et al. (2023). A valuable example is marteloscopes, i.e. finite forest stands with trees that are numbered, mapped and surveyed. Marteloscopes have been installed all over Europe in the past years, and training sessions are conducted by allowing students to walk through the physical forest and mark trees for cutting, identify possible habitat trees, or collect other relevant information (Krumm et al., 2019). Going one step further and visualizing the future stand development based on the decision taken would support the understanding and learning effect. Such visualizations are, however, not yet existing.

Such an application could theoretically be implemented either as an application that is interacted with on a screen, or as a virtual reality application (Murtiyoso et al., 2023). We chose the first approach on the one hand because the user needs less equipment, which facilitates the use of the application, and on the other hand because it can more easily be presented to a larger audience simultaneously.

4. Conclusions and outlook

In this paper, we have presented a software that can be used to visualize forests as they exist today, but also their potential future development depending on the applied silvicultural management strategies. This digital twin offers the possibility for users to experience forests in a digital environment. As a result, it could support forest planners when complex decisions are made, for instance by embedding the forest of interest in the surrounding landscape or zooming in and assessing how much light will reach the surface and which direction the light will come from after a planned intervention.

Moreover, the digital twin supports stakeholder communication by demonstrating the effect of harvesting operations. The opportunity to also look into the future development of the forest stand after the operations might be even more important. In the next decades, many forests will need to be adapted to handle the impacts of climate change. At the same time, more extreme disturbances are expected. In both cases it will be useful to equip foresters with a digital twin to visualize forest management, so that they can better explain and justify their ideas and decisions when communicating with stakeholders with different knowledge levels and objectives.

Notably, the digital twin presented here is a first version. We see several opportunities for further development in the future, all aiming at enhancing forest management.

- **Landscape visualization.** According to Zürcher et al. (2023), immersive media have great potential in understanding the properties and relationships of data obtained from a forest environment. Visualizing forest landscapes larger than the current size of 1 × 1 km

could better support regional change planning as well as stakeholder involvement in the planning process and might therefore expand the current focus on stand-level to a higher regional level.

- **More details of forests and their trees.** From the user's perspective, it would be attractive to add more details to the application to make the visualizations more realistic. This includes ground vegetation and more diverse tree models, e.g. more crown variations per species. This aspect is challenging because it reduces performance and detailed data about ground vegetation is rarely available.
- **Virtual reality.** Even though the benefits of immersive media are not fully proven (Bailenson 2018 in Zürcher et al., 2023), supporting head-mounted displays to enable a full virtual reality experience (in addition to a desktop version) might be attractive for many users. This would enable these users to fully immerse themselves into the virtual forest.
- **Forest growth simulators.** Adding bidirectional connections to forest growth simulators for flexible and quick testing of different management options. Currently, pre-defined scenarios are simulated by using a forest growth simulator. Next, simulator outputs are used as input for the digital twin and results are visualized using the virtual forest application. A bidirectional connection in combination with the possibility of marking trees for removal within the virtual forest application would make it possible to see the consequences of different harvesting strategies on the future development of the forest in real-time. This would be of high value for training areas like marteloscopes, in which the user can virtually cut trees or put them back on place.
- **Use of high-resolution data.** Visualizing of the present status of a forest, for instance by using LiDAR scans, to account for the increasing number of forest planners using LiDAR. In Switzerland, for instance, airborne laser scanning (ALS) data is provided by Swisstopo for free and is updated every 6 years. Making this data useable in such a digital twin would facilitate access and offer the option to benefit from this data. Moreover, we generally see the potential to develop it further with regard to advanced remote sensing technologies.
- **Use for other types of forests.** Currently, the application was developed to demonstrate the forest development under central-European growth- and management conditions. The use for other types of forests, such as tropical forests where plant diversity and understory cover are high, would also be possible. This would require 3D models of the occurring tree species and, in case of a high understory cover, could affect hardware requirements to still allow fluent movement in the virtual forest.
- **Training and monitoring.** The digital twin could be connected with the databases of national forest inventories to support the monitoring of plots over years.

CRedit authorship contribution statement

Stefan Holm: Conceptualization, Software, Visualization, Writing – original draft, Writing – review & editing. **Janine Schweier:** Conceptualization, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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